

		$\approx 1.28 \times 10^{-5} \text{ J}$
14	B	The internal energy of the ideal gas in the canister is independent of the gravitational potential energy (0.5 kJ) and the kinetic energy (1.0 kJ) of the gas. The internal energy of an ideal gas is only the kinetic energy due to the random motion of the molecules. As long as there is no heat transfer or work done on the ideal gas in the canister, the internal energy of the gas remains unchanged.
15	C	The potential energy of the particle undergoing SHM is maximum at $t = 0$